Communicating Qualitative Information: Charts, Diagrams, Maps, Photographs, and Clipart

6

> PowerPoint Jedi use their skills with PowerPoint to make difficult concepts clear. Me, I'm a PowerPoint Sith. I use my PowerPoint skills to confuse and obfuscate. If my boss doesn't have a glazed look of bewilderment, then the brief isn't complete. Embrace the dark side.
>
> —Mike Aitken (www.nbc-links.com/powerpoint.html)

If a perverse impulse leads you to want to confuse and obfuscate, it's easy to succeed. I've already mentioned a common technique, which is to pack immense amounts of text on every slide (using a tiny, barely discriminable font), from which you read random samples in a soft monotone. Even insomniacs will be overcome by such treatment.

Another sure-fire way to mow down an audience is to keep the material extremely abstract, so that it sounds good but the audience won't really know what you are talking about. Yet another approach that will sow bewilderment and confusion is to make claims that simply don't reflect reality—and assert them with great confidence. And you can use graphics that are meaningless or that have little bearing on your message; they might look very attractive, and distract the audience from your lack of clarity.

On the other hand, if your goal is to be clear and to the point, and to present a compelling case for your point of view, then you should present concrete evidence for your claims, which are presented in a logical sequence. To keep the audience's interest and attention, you need variety. Present graphics and sound to illustrate what you say—show and tell. This will not only make your presentation more interesting but will also make the material more comprehensible and will root what you say in reality.

The eight psychological principles apply to all communication graphics, including ones that convey qualitative information:

- *Charts* specify qualitative relationships among entities; family trees and flowcharts are charts.
- *Diagrams* are schematic pictures of objects or events that rely on conventionally defined symbols (such as arrows to indicate forces); "exploded" diagrams that show how parts of an object fit together and illustrations of football plays are diagrams.
- *Maps* present both qualitative and quantitative information; they not only portray geographical features, but also implicitly specify distances among locations, a quantitative variable.
- *Photographs and Clipart,* where the latter includes cartoons and drawings that illustrate people, ideas, or situations. These illustrations can be found online and easily pasted onto a slide (provided that they are in the public domain or proper copyright permissions are obtained; When in doubt, consult your lawyer!).

I begin with general recommendations, then turn to specific recommendations for each type of display.

General Recommendations for Charts, Diagrams, Maps, Photographs, and Clipart

The following recommendations apply to all of the types of graphics discussed in this chapter.

1. Include only graphics that help you make your point.

Decorations are distracting and may undermine your credibility if they seem cute or frivolous. Remember: A PowerPoint presentation should have a point!

2. Include only *the parts of* graphics that help you make your point.

Even when the graphic is appropriate, parts of it may not be. Include only material that is directly relevant to the specific point you are making at any given time. A diagram meant to show the driver's controls in an automobile may not be helpful to the viewer if many other automotive systems also are shown, as in "Don't" in Figure 6.1.

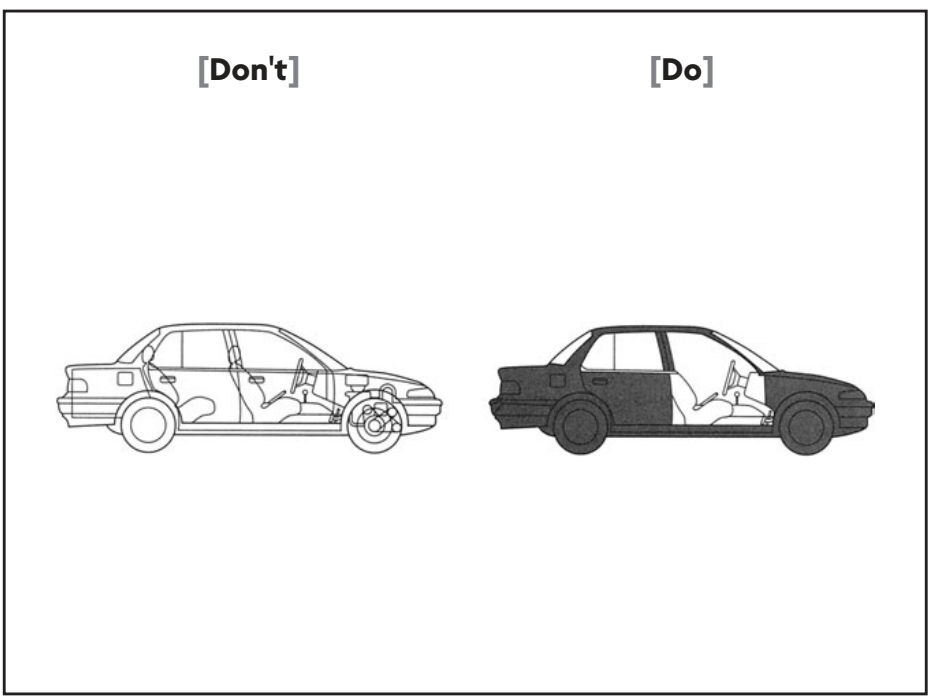

Figure 6.1. A diagram intended to show the driver's position relative to key controls is easier to read if the irrelevant components are eliminated and the relevant ones are highlighted.

3. Use symbols, concepts, and jargon that are familiar to the audience members.

Certain disciplines—electronics, genetics, and linguistics, to name a few—use specific specialized symbols, which should be used when

appropriate. Depending on context, however, the same symbol may have different meanings. For instance, an arrow may indicate parts to be matched, direction of force, or movement.

- If you even suspect that your audience may be in doubt about the meanings of the symbols you are using, state those meanings explicitly.

  Example: When describing a flowchart, indicate whether different shapes—such as squares and diamonds—have different meanings (such as an operation versus a decision/branching point).

- Also be sure that the audience is comfortable with the concepts and jargon you use. If you have any doubt, define the terms.

  Example: Instead of simply saying "a branching point" when describing a flowchart, say "A branching point, where a decision is made about the next sequence of steps to take."

4. Ensure that all aspects of the graphic are clearly discriminable from the background.

The graphic must have enough contrast with the background to be visible throughout the room and must be large enough to be legible even from the back row.

5. Change visual or auditory characteristics only to signal a change in information.

The Principle of Informative Changes applies to all types of graphics.

Example: Clipart or a photograph alerts the audience to pay attention to a new idea or claim; if nothing is new, don't introduce a new piece of Clipart or a photo.

6. Organize the components of complex displays.

A graphic can effectively convey a complex set of qualitative relations if the complexity doesn't visually overwhelm the relations depicted.

To avoid producing a tangled web or overwhelming pastiche, divide different aspects of the display into separate perceptual units (see Figure 6.2). These units can be defined by:

- Similar shapes
- Similar colors
- Similar shadings
- Similar sizes
- Similar line weights
- A line or arrow that links objects
- Nearby locations, which are clearly distinct from other clusters

Finally, you can also produce groups simply by presenting a subset of the material at a time, and build up the whole display over the course of a series of slides.

In "Do" in Figure 6.2, the phases of the data collection and evaluation process are distinguished from the decision processes. Similarly, many flowcharts use diamond shapes when a decision is made, squares when a process occurs, and circles when input or output occurs.

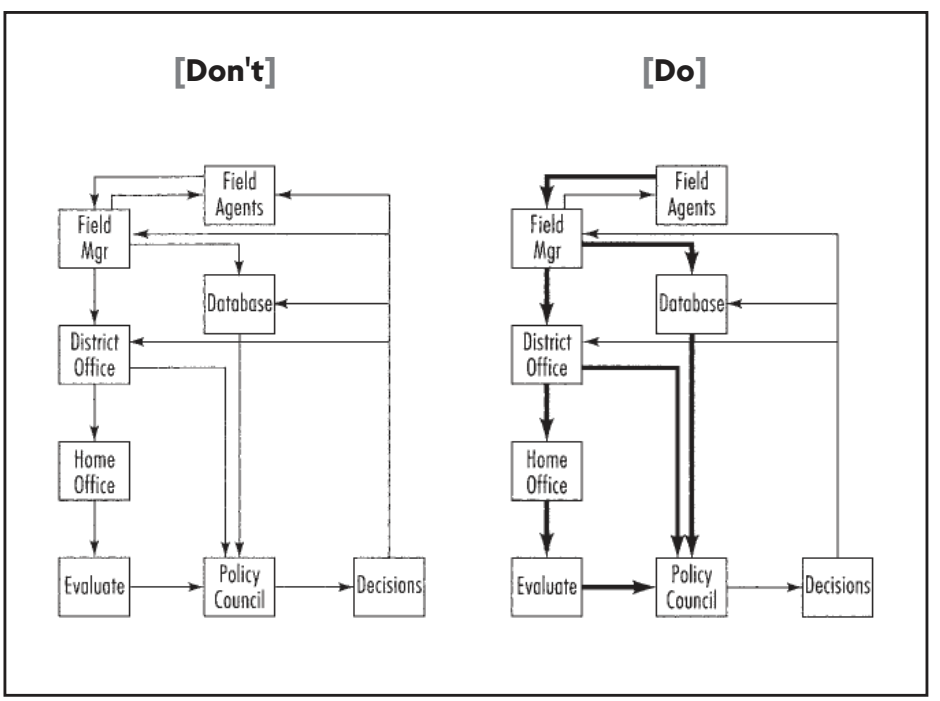

Figure 6.2. A complex chart can be sorted out relatively easily if it is visually organized into components.

7. Respect the *Rule of Four*: Don't expect viewers to attend to more than four perceptual units on a slide.

If you want viewers to pay attention to more than four objects in your graphic, devise a way to present them as groups (using the variables noted in the previous recommendation), so that there

are no more than four groups. The crucial limiting factor is the number of perceptual units (i.e., groups), not the actual number of objects.

How can you tell how many perceptual units are present? Fortunately, studies have shown that people are remarkably accurate and consistent at drawing lines around the parts of a displayed object or scene, which correspond well to the perceptual units. Thus, you need only consider where you would draw lines between parts or groups of objects, and you'll have a good sense of how many units there are.

8. Use animation and vary salience to guide the audience through the display.

If the point of the graphic requires the audience to grasp a sequence of events, use animation to illustrate that sequence. If you need to vary the speed of animation, use the "custom animation" feature (under the Slide Show menu item, at least in my current version of the program), and put a set of objects on a single slide, defining the animation values for each. Alternatively, make a series of slides, like frames in a film, and use the "slide transition" feature to specify how long each one is visible (but in this case you will need to experiment with the number of frames and exposures in order to produce acceptable motion). If the display does not lend itself to movement, vary salience—by changing the display over time—to guide the viewers' attention.

Example: By systematically changing the colors of specific states of the United States in a series of slides, you could illustrate the course of the western expansion during the 18th and 19th centuries.

9. Use multiple panels to highlight specific comparisons.

If your initial display is too complicated, and you cannot easily group the elements into 4 or fewer perceptual units (as noted above), break the display into separate panels.

- Decide which units are most closely interacting (e.g., in a flowchart) or are most closely related (e.g., adjacent territories in a map), and present them together.
- To provide the overall context, gray out the rest of the display.
- If the display is so large or complex that individual components cannot be clearly seen, "zoom in" on one part at a time. Provide a

"thumbnail sketch" at the upper right, showing (e.g., with a red square) the relative location of the current material within the context of the whole. In such cases work through the display systematically, to help viewers integrate each portion into the whole.

Charts: 6 Pointers

Charts can make complex sets of information clear at a glance, or can make complex information seem even more complex. For instance, consider one chart that was created in the wake of Hurricane Katrina. This hurricane was no laughing matter, nor was the feeble response of the Federal Emergency Management Agency (FEMA). Nevertheless, Jon Stewart, the host of the *Daily Show* (broadcast on Comedy Central), couldn't help commenting on a chart released by FEMA (during the week of 26 September 2005), Figure 6.3. He said: "What should FEMA have done? Perhaps the answer can be found on their website. . . . This chart [shown below], clearly depicting the agency's responsibilities in the event of a disaster. . . . It begins with a response to a disaster, leads to recovery, mitigation, risk reduction, prevention, preparedness . . . (dramatic pause) and

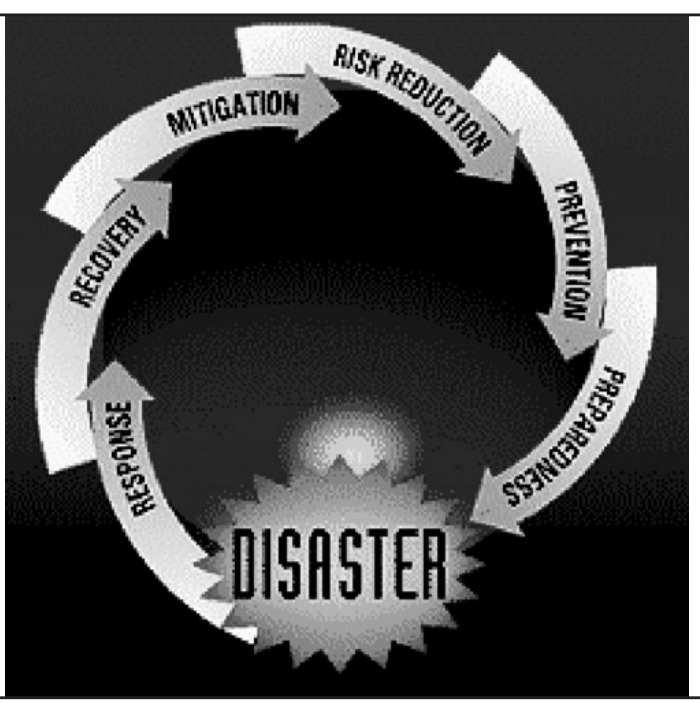

Figure 6.3. The actual chart from the FEMA website. Can you understand what it means?

ends up BACK IN DISASTER! . . . In truth, FEMA did exactly what they *said* they were going to do" (http://presentationzen.blogs.com/presentationzen/2005/10/fema_chart_beco.html).

Charts are visual displays that arrange information into categories or structures. Usually, lines connect the elements of a chart to indicate structure; but the arrangement of the elements (such as boxes or even just words) on the screen may be enough to show the relationship clearly. The FEMA Chart? The bad news is that it inadvertently shows how disasters beget disasters. Moreover, according to their website, their cycle actually begins with (not ends with, as shown here), "Preparedness" and ends with "Response," but this still doesn't fit with the use of arrows—which might suggest that the "Response" leads to the disaster! The good news is that it's almost impossible to read this chart (notice how hard it is to read all upper case, especially when the words are at odd angles and even upside down).

1. Use a chart to convey overall organizational structure.

Using a chart to illustrate the structure of an object, organization, or event takes advantage of the Principle of Compatibility. Relationships such as "is a member of," "follows," "works for," and "is descended from" can be illustrated clearly by the spatial relations of the entities in the display.

2. Use a chart to illustrate sequences of steps over time.

The PowerPoint program lets you show even complex sequences of events by building up a chart, a part at a time.

3. Use a list to convey one relation among several entities.

The alternative to a chart is a list, which is preferable if only a single relation and four or fewer entities need be considered.

Example: A simple list can effectively convey a simple organization:

Company Divisions
- Consulting
- Publishing

People can read such a list faster than they can decipher a chart, and a short list will tax short-term memory less than would a multi-element display. However, if you want the viewers to compare numerous different combinations of entities (such as the relations

between various managers and their superiors in the company), a chart is better than a list. In the wise words of Albert Einstein, "Everything should be made as simple as possible, but not one bit simpler."

4. Show more inclusive categories higher in the display.

Say that the problem is to show how the Mega Corporation breaks down into separate divisions. Place the larger entities higher in a chart with the entities that are contained within the larger entities depicted beneath them.

Example: Mega Corp. has automotive, consumer electronics, and ranching divisions. These would be symbolized in a chart by boxes (or pictures of representative objects) in a row directly beneath the box (or picture) representing the company. Beneath each of these three entities would be boxes or pictures for their components.

5. Use a layout compatible with the subject matter.

A chart detailing the command structure of an organization (Figure 6.4b) should start at the top and work down, as in "Do"; it should not go from left to right.

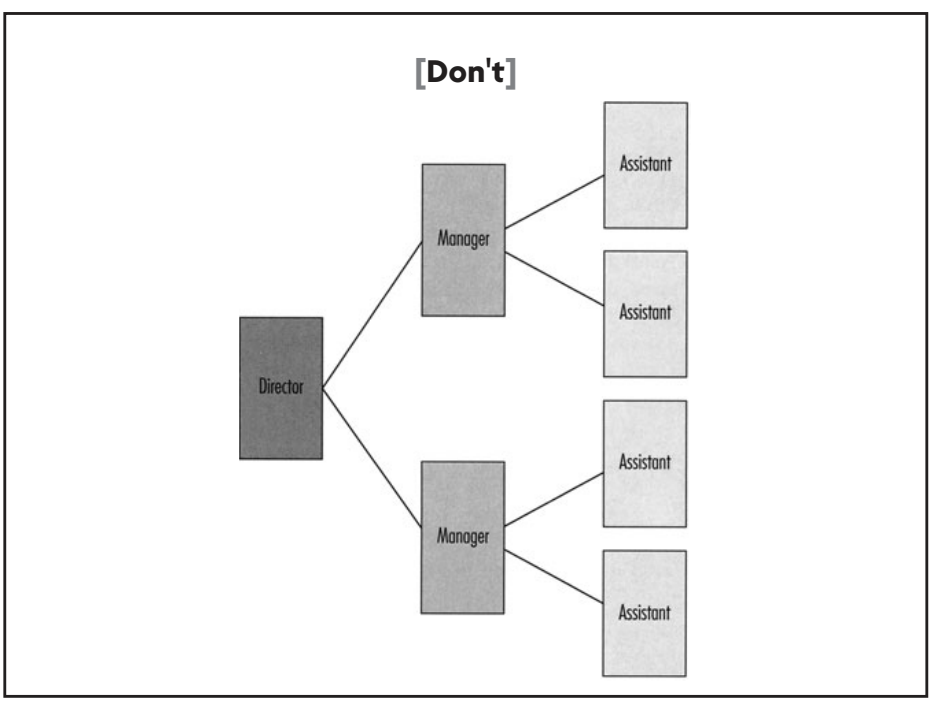

Figure 6.4a. Category inclusion and command structure both correspond to vertical spatial relations because some categories or positions are "over" others; the unfolding of events over time is shown better in a horizontal layout.

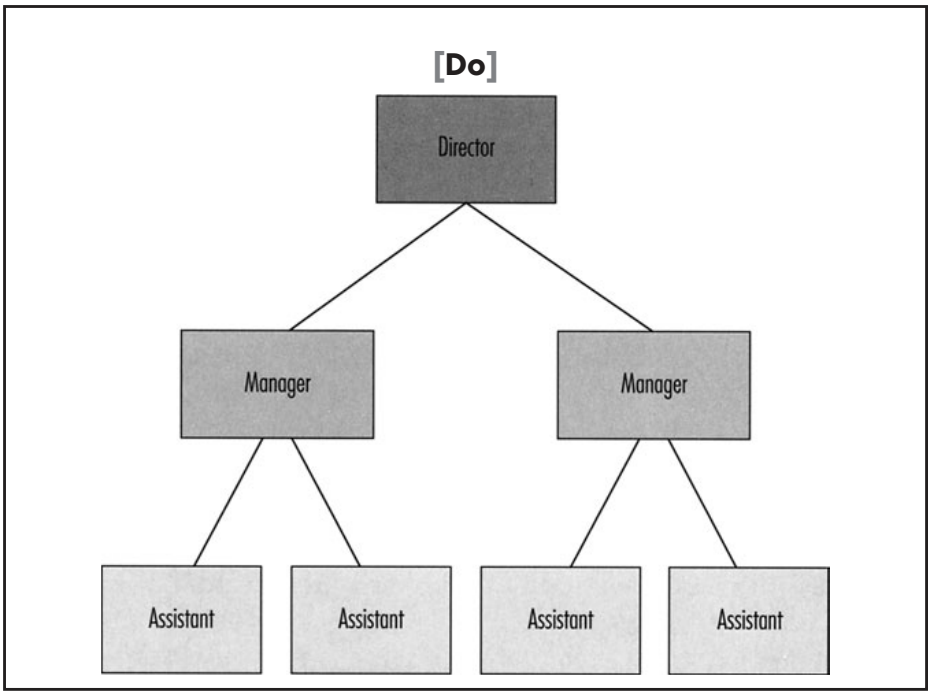

Figure 6.4b.

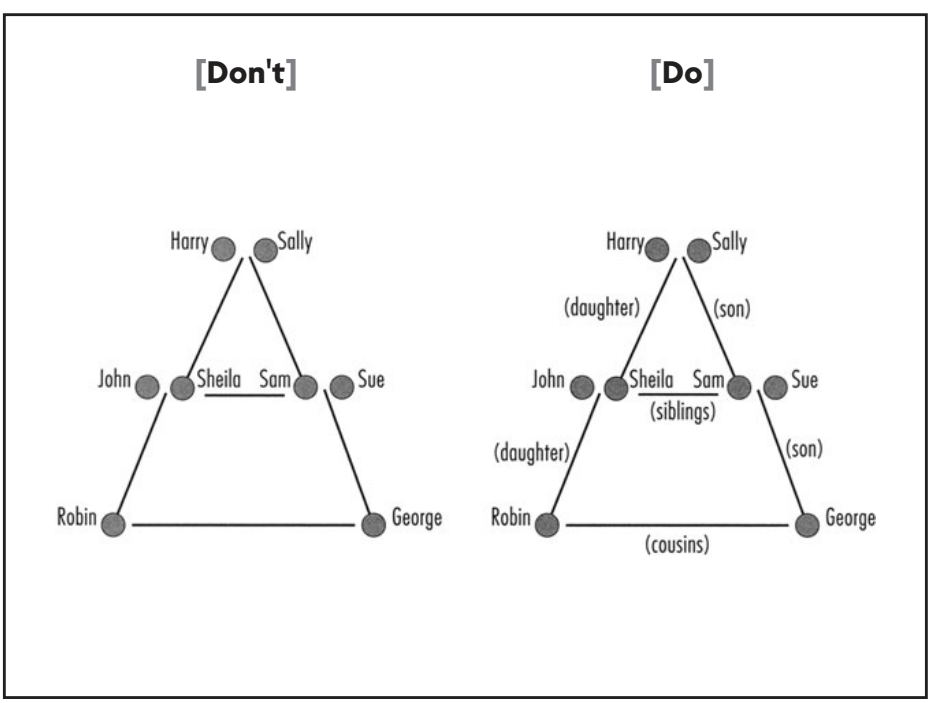

Figure 6.5. Every important piece of information, whether entity or relationship, should be identified in the display.

- All else being equal, respect the convention in Western culture and make a sequence over time proceed from left to right.
- However, respect the specific conventions of your audience. For example, computer programmers have developed a specialized subculture, where flowcharts are often presented from the top down.

6. Identify relationships.

Identify every important piece of information in a display, as in the "Do" version of this family tree. (Figure 6.5).

Diagrams: 8 Recommendations

Diagrams are pictures of objects or events that use both pictorial elements and symbols (such as arrows to show movement) to convey information—to show the wiring in your kitchen, the way the Eiffel Tower sways in a strong wind, or the assembly of your new barbeque. Unlike charts, parts of diagrams must resemble the things they represent.

1. Use a diagram to illustrate the structure of an object or event for a specific purpose.

When designing a diagram to illustrate the structure of an object or event, keep the purpose of the diagram in mind from the outset. If you don't, you may end up including too much or too little material—leaving the viewers frustrated or annoyed.

2. Explode a diagram to emphasize the shapes or spatial relations of individual parts.

If the parts are partially concealed and you want to make their shapes or spatial relations very clear, move parts away from the whole—but be sure to use an arrow or other visual means to indicate clearly where each part belongs.

3. Don't explode an object too widely for easy recognition.

If the parts are exploded in order to show how they fit together, don't expect the viewer to extrapolate or remember precise spatial relations

among parts of a display; the brain processes shape and spatial relations in separate systems and does not easily combine the two very accurately. In an exploded diagram, show parts near their actual locations on or in the object depicted, as in "Do" in Figure 6.6.

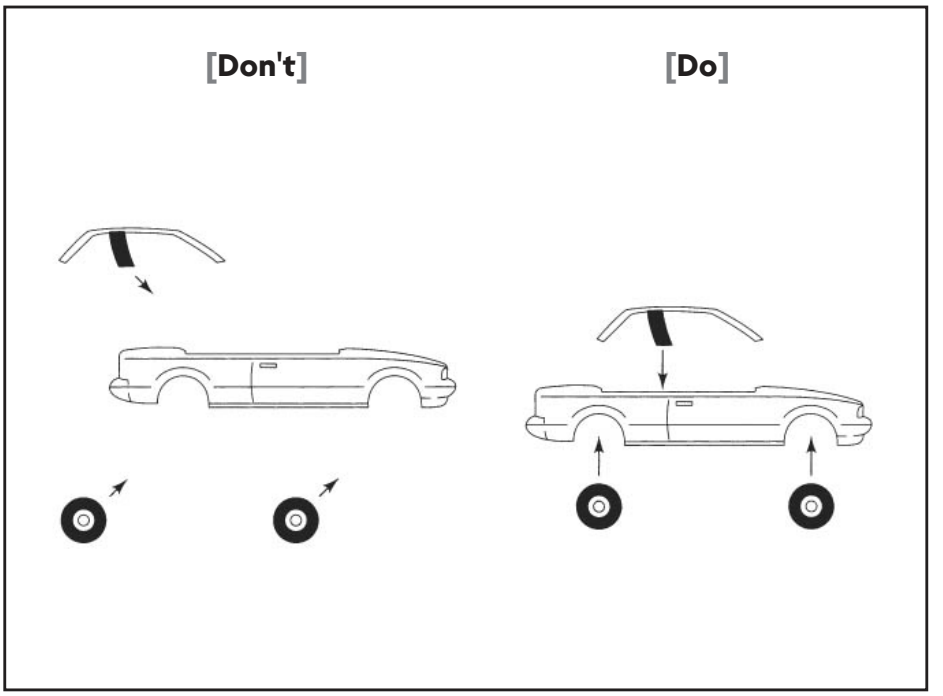

Figure 6.6. If spatial relations are distorted to show components, the distortion should be easy to reintegrate; otherwise, readers will have to expend effort to see the components in place.

4. Ensure that different components of the diagram are organized into separate perceptual units.

Exploding is not the only way to specify the shapes of parts; different colors or shading can also be used to individuate parts. If distinct parts are to be shown, ensure that they are easily identifiable as distinct parts.

5. Use compatible movement when presenting a diagram of a dynamic event.

You can simulate the process of assembling or modifying an object or tracking a process (such as the effects of a strong wind hitting the side of a skyscraper) by using animation to add, remove, or modify parts of a diagram over time.

- Ensure that the order and way in which parts are added, removed, or modified (e.g., shown to bend) mimics the actual order or way in which they would be added, removed, or modified. This technique allows you to discuss each relevant part before turning to the next, and the use of animation will draw the viewers' attention to the next topic of discussion.

6. Choose a point of view in which all critical aspects of the material are visible.

Information that is hidden may as well not exist.

7. Show all parts of static diagrams from the same viewpoint.

The cubists shocked and befuddled their contemporaries by painting objects as if they were seen from multiple points of view simultaneously. At the time, this was a good way to get the attention of the critics and the public, and to lead viewers to consider reality in new ways. But what works well in art (at least for some people) does not necessarily work well in communication graphics. In fact, most people will be confused and have to struggle to understand what you are showing them if you show it from more than a single point of view: The brain assumes that an object is seen from only a single vantage point, and effort will be required to reorganize the display mentally if it is not.

- Similarly, never have one part of the display drawn in two dimensions with a three-dimensional part tacked on to it.

8. Use animation to illustrate three-dimensional structure.

One way to illustrate the structure of a three-dimensional object is to use animation to rotate it in depth. The rotation will be most effective if:

- It is as slow as possible while still revealing the overall structure.
- Four or fewer perceptual units are highlighted (e.g., with a distinctive color).

Example: I've seen many rotating images of the human brain, with colors highlighting specific parts—such as the regions involved in vision. Sometimes the colors come on in sequence,

illustrating how the specific brain areas interact. Such information is much easier to grasp when the object appears three dimensional, which it does when rotating.

Maps: 12 Key Considerations

The following is reputed to be an actual transcript of a radio conversation between a Canadian communications officer and a U.S. naval vessel, recorded off the coast of Newfoundland in October 1995:

Americans: Please divert your course 15 degrees to the north to avoid a collision.
Canadians: Recommend you divert YOUR course 15 degrees to the south to avoid a collision.
Americans: This is the Captain of a U.S. Navy ship. I say again, divert your course.
Canadians: No. I say again, you divert your course.
Americans: This is the aircraft carrier USS Lincoln, the second largest ship in the United States' Atlantic fleet! We are accompanied by three destroyers, three cruisers, and numerous support vessels. I demand that you change your course 15 degrees north, that's one-five degrees north, or countermeasures will be undertaken to ensure the safety of this ship!
Canadians: We are a lighthouse. Your call.

If we are to accept this report as gospel (which may be dubious, given that it is posted on the Web both as a joke and as an "actual transcript," as was characterized by the then-president of Duke University, Nannerl O. Keohane, in a speech to the Canadian Club of Ottawa on 28 October 1999, http://canada.usembassy.gov/content/textonly.asp?section=can_usa&document=keohane&subsection1=moreinfo), you could easily fault the participants for their attitudes or communication skills. But in my view the major culprit would probably have been something else: the navigation maps. These maps should have made the location of the lighthouse crystal clear, and should have made it easy to establish where the ships were in relation to the lighthouse.

Maps are drawings that function as pictures of a physical layout. As stylized pictures of a territory, maps provide information about locations and relations among them (usually in terms of relative distances and routes between them). By using conventional markings, they can also provide quantitative information about various locations, such as average temperature, voting patterns, and population distribution.

1. Use a map to show distances of segments or angles between segments of a route.

Use a map if the relative distances between landmarks or angles of turns are important.

2. Use a map if more than one route is possible.

One alternative to a map is a verbal description of a route or territory. However, if you can reach the destination from a number of different starting places or by more than one route between two locations, a map is distinctly more helpful.

3. Use a map to label complex sets of information about a territory.

Maps are also a convenient way to provide various sorts of information about sets of localities; a map allows the viewer to see the relations among different values in many locations.

Example: If population is presented (perhaps by bars standing on particular locations), the viewer can not only see the relation of population to specific landmarks, such as rivers or the sea, but also gain an impression of the pattern of population variations as a whole.

4. Provide neither more nor less detail than required for the point to be made.

If you want to illustrate how to get to the nearest fire station from Harvard Square, it would not be helpful to include every road in the region, as in "Don't" in Figure 6.7. Instead, illustrate the roads that are efficient routes between the two locations, providing enough landmarks that the traveler won't get lost.

- Moreover, if distance per se is not crucial but the sequence of turns is, there is no reason to draw the map to scale. However, if you don't draw the map to scale, indicate when distance has been compressed—such as by using the same sort of truncation marks used to indicate that a Y axis in a graph as been altered (which has the advantage of probably being a familiar symbol to most viewers). However, because some viewers may not be familiar with

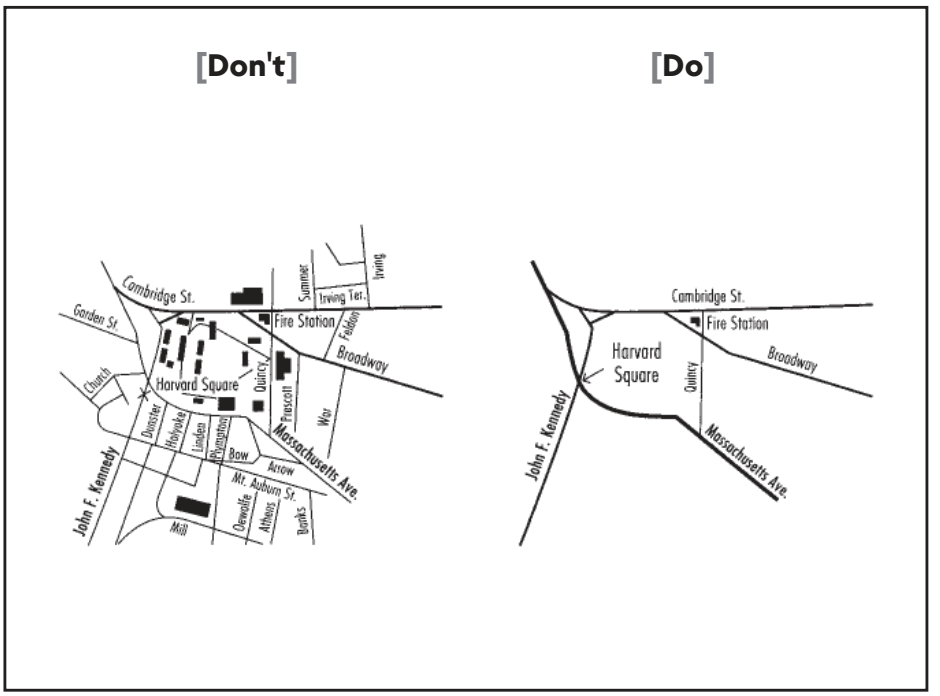

Figure 6.7. A map that includes too much extraneous detail for its purpose is not helpful; to specify the location of the firehouse, only the efficient routes are necessary.

the conventions for indicating truncation, be sure to explain what such symbols mean.

5. Don't vary sizes of regions to convey precise quantitative information.

Some years ago it became fashionable to design maps of the United States in which the sizes of the states were varied, as in "Don't" in Figure 6.8, to show relative rates or quantities—per capita beer consumption, number of murders, and so forth; the more per the state, the larger the area. At best, such maps convey only a rough impression of rank ordering, and fall short if actual amounts or even precise ordering is to be conveyed. We humans simply are not very good at estimating area and have trouble comparing relative areas of differently shaped regions.

- A better way to convey amount is to draw a bar, stacked set of icons, or similar symbols at each location, varying length or height to convey quantity; our visual system can compare relative lengths well.

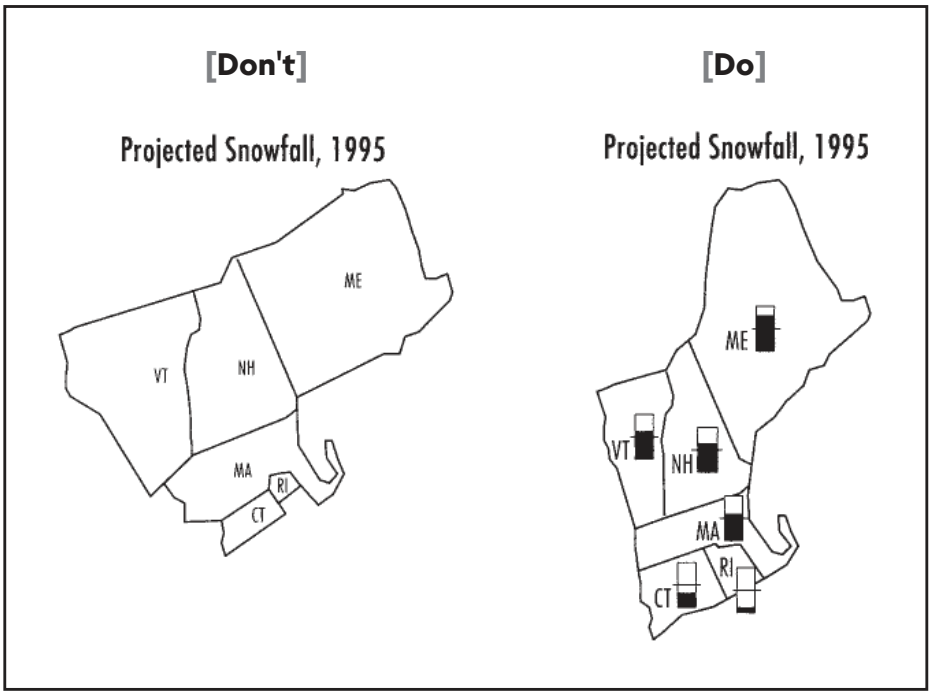

Figure 6.8. It is almost impossible to extract relative amounts from differently shaped areas, even if they are familiar and not distorted. Vary extent along a line or bar instead (the framework around each bar and the short horizontal lines are designed to help viewers compare the black bars).

6. Ensure that shapes of meaningful regions are easily identifiable.

Another problem with varying the size of a region to convey information is that the shape may become unrecognizable. If Maine is made larger, its shape may have to be modified to fit it against its neighbors, as happened in the "Don't" example in Figure 6.8.

7. Don't vary height and width of location markers to specify different types of information.

The map in "Don't" in figure 6.9 uses the height of the markers to indicate the population and the width of the markers to indicate the mean temperature of each location. But our brains register the area of the markers, not each dimension separately. Don't vary a shape's height and width (or—as discussed in Chapter 4—a color's hue, lightness, and saturation) to specify values of distinct dimensions.

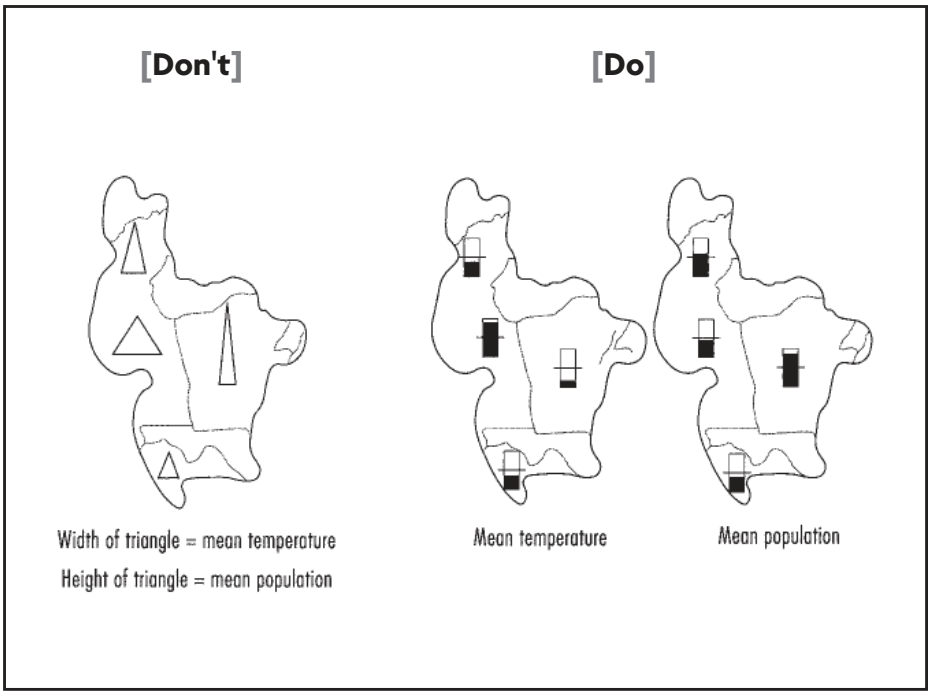

Figure 6.9. If both the width and height of the triangles are varied, we see neither variable very well; it is far better to use two different displays, one for each variable.

8. Avoid visual illusions that distort distance and direction.

Visual illusions affect our perception of distances. Vertical lines appear longer than horizontal ones, and straight lines that are interrupted, as in "Don't" in Figure 6.10, appear to be displaced; the diagonal road passing under the highway is actually straight, but it appears to jog under the overpass. This is not a problem in "Do".

9. If distance is important, use grid markings.

Some maps are intended to indicate the relative locations of regions, to show the locations of specific routes, or to provide other kinds of qualitative information. Other maps, however, specify the actual distances between locations. In these cases, inner grids are useful if

- grid lines are relatively thin and light (this recommendation is violated in "Don't" in Figure 6.11).
- more tightly spaced grid lines occur when greater precision is required.
- heavier grid lines are inserted at equal intervals, to help viewers search.
- inner grid lines pass behind the lines, bars, or other symbols.

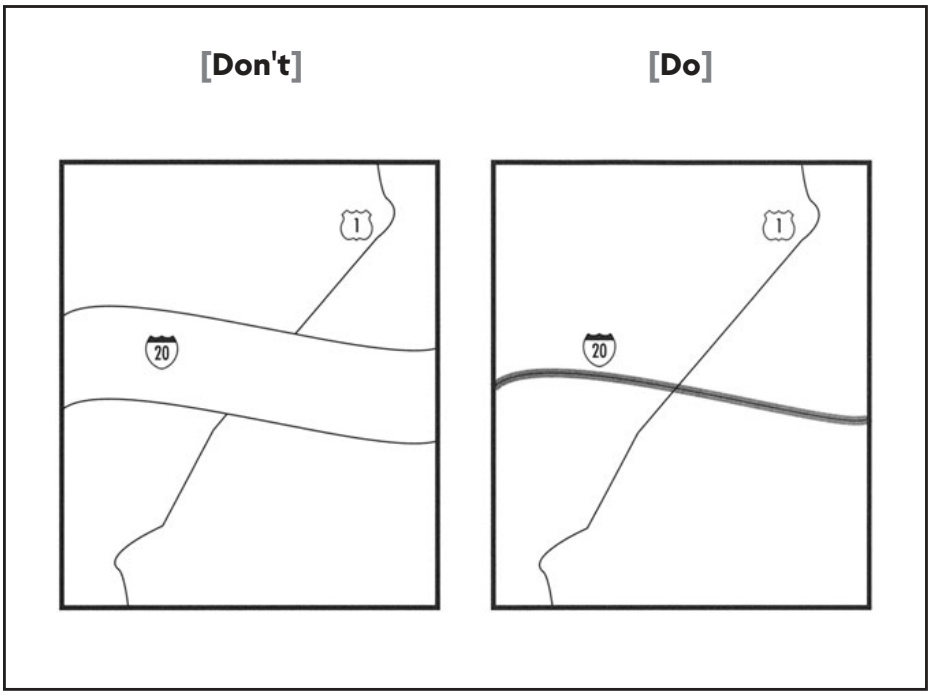

Figure 6.10. Beware of visual illusions (including exaggerated distances of vertical lines) when producing maps. In the map on the left, Route 1 seems to jog as it passes under the Interstate, but in fact it does not.

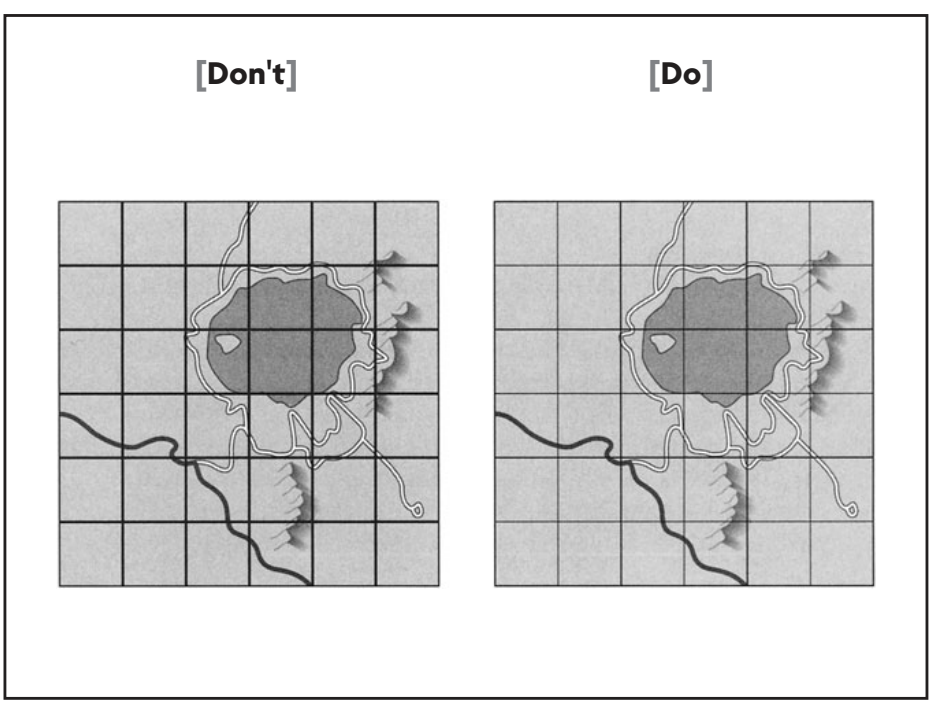

Figure 6.11. If precise distance is not important, there is no reason to include grid lines; if grid lines are included, they should not obscure the features of the territory.

10. Make the scale equal one or more grid units.

The scale of a map is a kind of key, pairing a line length with a unit of distance, which the viewer uses to estimate the distances between the locations represented on the map. To be effective, the scale should correspond to one or more of the units used to produce the grid lines; if it does not, the viewer will have to reorganize the unit line length, mentally dividing it into the corresponding lengths on the grid—and this operation is neither efficient nor accurate.

- If the grid lines are spaced every half inch on the map, the scale should indicate the distance in terms of this unit length or in terms of a familiar multiple of this increment, say, 1 inch. It's often most useful if the scale indicates a unit that corresponds to the distance between heavy grid lines as in "Do" in Figure 6.12.

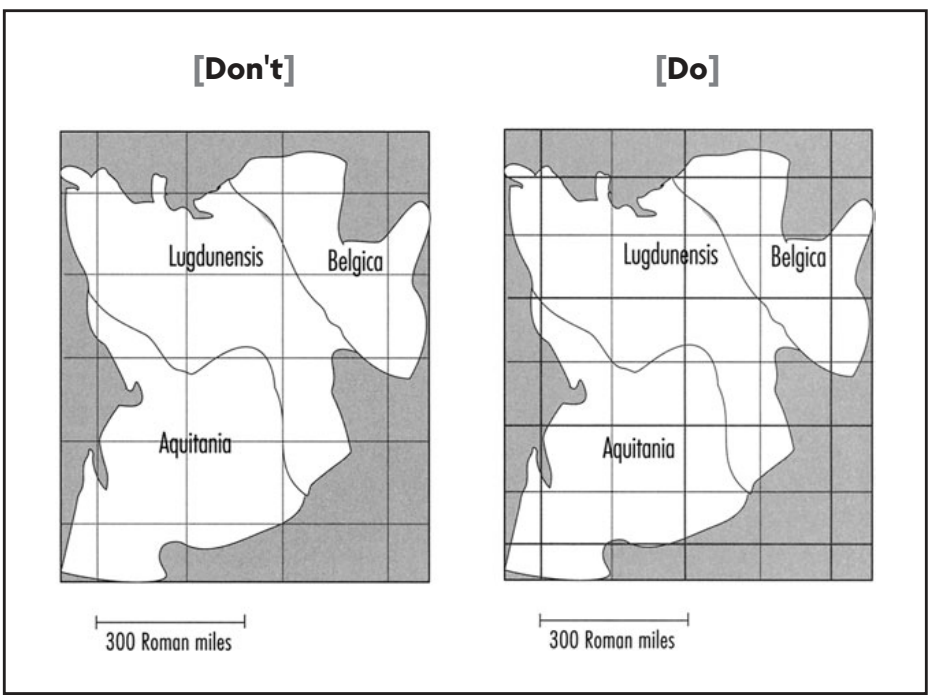

Figure 6.12. If the scale does not line up with the increments of the grids, mental division is necessary to use it; spare the viewer this effort by aligning the scale and grid increments.

11. Make more important routes more salient.

Use the most important routes as the backbone of the map, as in "Do," in Figure 6.13, allowing other routes to be organized by them. This can be accomplished simply by making the more important routes more visually distinctive, using color or line weight.

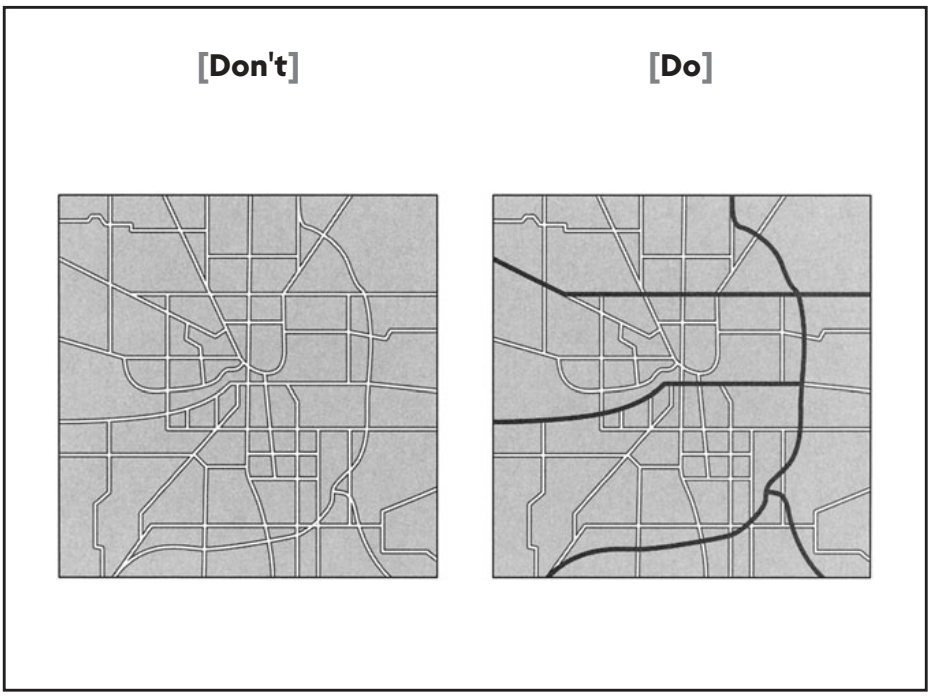

Figure 6.13. Making the major roads correspond to more salient lines will help viewers to organize the map as well as to find the most efficient routes.

12. Label important distances directly.

If you want the audience to know the distances between specific pairs of locations, draw lines between them and label the distances directly (the lines should indicate routes).

Photographs and Clipart: 10 Recommendations

Photographs and Clipart are perhaps the most insidious types of illustrations, which practically invite abuse. Many of us can't help but succumb to the temptation to include that cute photo, cartoon, or drawing—and we end up distracting the audience and undermining our message. Nevertheless, photos and clipart do have their place.

1. Use photos and Clipart to define the context.

Clipart can set the stage for your entire presentation or a portion of your presentation, or it can help you make a specific point.

Example: A drawing or cartoon of historical figures, such as Benjamin Franklin, can set the stage for a contemporary treatment of a relevant topic—such as contemporary methods of making bifocal lenses.

2. Use photos and clipart to introduce an abstract idea.

Sometimes very abstract ideas have become associated with concrete icons. Such illustrations can put the audience in a frame of mind to organize what you have to say in terms they already know—which is always a good idea.

Example: Albert Einstein's visage has become a symbol for genius, and the Statue of Liberty has become a symbol of freedom for immigrants.

3. Use photos and Clipart to evoke a specific emotion.

In some situations, you may want your audience to feel a specific emotion. This can be part and parcel of an argument you are making. Research in cognitive neuroscience has shown that many decisions are driven by emotional associations, and thus an important part of any argument is to create the appropriate emotional reactions to points being made.

Example: If you want to argue that additional funds should be allocated for helping the homeless, you might show a series of photos of dispirited and dejected homeless men, women, and children living on the streets of various cities.

4. Use photos to present evidence.

Even in these days of computer graphics, many people still react as if "seeing is believing." Thus, use photos to present evidence for your case. Moreover, presenting both text and a graphic will greatly help your audience retain the information.

Example: If you claim that more people drink tea than coffee worldwide, present a montage of photos of people from different parts of the world drinking tea.

5. Use photos and Clipart to give the audience time to "come up for air".

Photos and Clipart can also be useful as a break in the steady flow of information, allowing the audience a moment to reflect and digest. This is especially the case if the photo or Clipart is humorous.

- If you punctuate your presentation with such graphics, be sure that they are relevant to the ongoing thread of your discussion.
- If they are not, they will derail the viewers and disrupt the line of your argument.

6. Use photos and Clipart to direct the audience's attention.

Because attention is drawn to novel stimuli, if you've been presenting mostly text and then present an illustration, the audience will automatically look at the illustration.

- You can use this to your advantage to introduce a new module of the presentation, or to give the audience "time to come up for air."
- But keep in mind that when their attention is drawn to the illustration, the audience won't read text or listen fully to what you say—so, give them a moment or two to absorb the illustration before going on.

7. Choose photos and Clipart that are representative of your point.

If you use photos or Clipart to illustrate a specific situation, select an illustration of an average, typical example of the type of object or situation.

Example: If you want to discuss the proportion of different sorts of animals that live among humans in urban areas, use an illustration of a pigeon-like bird to illustrate "bird," not a crow or woodpecker.

8. Ensure that the style of photos or Clipart is compatible with your message.

People not only grasp the meaning of photos and Clipart but also register the visual qualities of the illustrations themselves. Ensure that the medium meshes with the message.

Example: Don't use a cute cartoon to illustrate a serious point, and don't use a black-and-white photo to illustrate high-technology.

9. Ensure that illustrations face the center of the slide.

Objects in illustrations typically have a direction; a face looks to the right or left, a car points in one direction or the other, and so on. The orientation of the object will direct the viewers' attention. If you have a face on the right side of the slide looking off to the right, the viewers will have a tendency to do the same—and thus be looking off-screen, rather than where you want them to be focusing when the next slide appears.

10. Ensure that photos and Clipart do not become too grainy when inserted into the slide.

Clipart is often drawn from other sources, such as websites (but, again, keep copyright issues in mind when you obtain such downloads). The resolution of the art can vary dramatically, and you should be sure that it does not become grainy if you need to expand the art to make it visible from the rear of the room. Not only does grainy art look bad (and make you look bad by extension), but it also can require effort to understand—never a good idea.

From the Principles to the Point

As usual, these recommendations are based directly the eight psychological principles. I again revisit the principles and note which ones contributed primarily to each of the major recommendations I offer.

Principle of Relevance: Communication is most effective when neither too much nor too little information is presented. (We understand and remember a message more easily when the right amount of detail is used to make the point.)

- Include only graphics that help you make your point.
- Include only the parts of graphics that help you make your point.
- Use a list to convey one relation among several entities.

- Choose a point of view in which all critical aspects of the material are visible.
- Provide neither more nor less detail than required to make the point.

Principle of Appropriate Knowledge: Communication requires prior knowledge of relevant concepts, jargon, and symbols. (We understand and remember a message more easily if it connects to what we already know.)

- Use symbols, concepts, and jargon that are familiar to the audience members.

Principle of Salience: Attention is drawn to large perceptible differences. (Big relative differences grab attention.)

- Use animation and vary salience to guide the audience through the display.
- Explode a diagram to emphasize the shapes or spatial relations of individual parts.
- Make more important routes more salient.
- Use photos and Clipart to direct the audience's attention.

Principle of Discriminability: Two properties must differ by a large enough proportion or they will not be distinguished. (We need contrast to distinguish shapes, colors, or positions from each other and from the background.)

- Ensure that all aspects of the graphic are clearly discriminable from the background.
- Ensure that shapes of meaningful regions are easily identifiable.
- Ensure that photos and clipart do not become too grainy when inserted into the slide.

Principle of Perceptual Organization: People automatically group elements into units, which they then attend to and remember. (These groups are easier to see and remember than the isolated components would be.)

- Organize the components of complex displays.
- Ensure that different components of a diagram are organized into separate perceptual units.
- Use animation to illustrate three-dimensional structure.
- Don't vary height and width of location markers to specify different types of information.
- Don't vary the sizes of regions to convey precise quantitative information.

Principle of Compatibility: A message is easiest to understand if its form is compatible with its meaning. (For better or worse, the mind tends to judge a book by its cover.)

- Use a chart to convey overall organizational structure.
- Use a chart to illustrate sequences of steps over time.
- Show more inclusive categories higher in the display.
- Use a layout compatible with the subject matter.
- Use a diagram to illustrate the structure of an object or event for a specific purpose.
- Use compatible movement when presenting a diagram of a dynamic event.
- Use a map to show distances of segments or angles between segments of a route.
- Use a map if more than one route is possible.
- Avoid visual illusions that distort distance and direction.
- Use photos and Clipart to define the context.
- Use photos and Clipart to introduce an abstract idea.
- Use photos and Clipart to evoke a specific emotion.
- Choose photos and Clipart that are representative of your point.
- Ensure that the style of photos or Clipart is compatible with your message.
- Ensure that illustrations face the center of the slide.

Principle of Informative Changes: People expect changes in properties to carry information. (And we expect every necessary piece of information to be indicated by a change in a perceptible property.)

- Change visual or auditory characteristics only to signal a change in information.
- Identify relationships.
- Show all parts of static diagrams from the same viewpoint.
- Label important distances directly.
- Use a map to label complex sets of information about a territory.

Principle of Capacity Limitations: People have a limited capacity to retain and to process information, and so will not understand a message if too much information must be retained or processed. (From a communicative point of view, less can be more!)

- Respect the *Rule of Four:* Don't expect viewers to attend to more than four perceptual units on a slide.
- Use multiple panels to highlight specific comparisons.
- Don't explode an object too widely for recognition.

- If distance is important, use grid markings.
- Make the scale equal to one or more grid units.
- Use photos to present evidence.
- Use photos and Clipart to give the audience time to "come up for air."